NucAE: reconstructing nucleosome coverage from sparse cfDNA sequencing

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1 What the system does

Cell-free DNA fragments in blood are protected from degradation by nucleosomes, so the midpoint of a fragment marks approximately where a nucleosome sat. Counting those midpoints along the genome gives a coverage track whose peaks are nucleosome positions.

At high sequencing depth this track is clean. At the depth a clinical assay can afford it is mostly noise. The model takes the noisy track and reconstructs what the deep one would have looked like.

	input	**target**
coverage	∼5×	full depth
window	25,000 bp	25,000 bp
extra	reference sequence	—

The reconstruction is judged against the deep track that was measured on the same sample, so the ground truth is real data and not a simulation.

2 Input

The pipeline begins from **fragment centre counts**, one file per chromosome:

```
chr8  4103877  3  2.8714
```

Four tab-separated columns: chromosome, 1-based position, raw count, GC-corrected count. Only covered positions are written; an absent position means zero coverage. The pipeline reads the GC-corrected column.

These files are produced from BAMs by a separate project (`cfdna_tfp_profiling`). That boundary is deliberate: this repository starts from counts on disk and never reads a BAM.

The undersampled arm is made by randomly subsampling the BAM by read name, so both ends of a fragment are kept or dropped together — a half fragment has no midpoint. The seed and the fraction are recorded, since they are the only things that make that subsample reproducible.

Two further inputs describe the genome rather than the sample: the reference sequence, and a **mappability list** of regions where short reads can be placed reliably.

3 Preprocessing

3.1 The smoothing chain

Raw midpoint counts are spiky. Four steps in fixed order turn them into a nucleosome track:

1. **Blacklist zeroing.** Positions in unreliable regions are set to zero.
2. **Whittaker–Eilers smoothing** ($\lambda = 1000$, order 2). Fits a smooth curve through the counts.
3. **Gaussian smoothing** ($\sigma = 30$ bp). Blurs at roughly the scale of a nucleosome.
4. **Subtract a running median** (window 375 bp). Removes slow drift in overall depth, leaving the nucleosome oscillation centred on zero.

Both coverage levels go through the identical chain. Parts of it are known to be redundant, and it is kept unchanged so that results stay comparable to earlier work.

3.2 Smoothing happens when the data is read, not on disk

The stored file holds *raw* counts. Smoothing is applied by the dataloader each time a window is read.

This costs training time — it is the single largest cost per window — and buys two things: smoothing is destructive and some of its parameters are still open questions, and one file can serve consumers that want the data differently.

3.3 Windows and the halo

The genome is cut into non-overlapping 25,000 bp windows, tiled from position 0. Incomplete windows at a chromosome's end are dropped, because the model takes a fixed input length.

Smoothing a window in isolation would corrupt its edges, since the filters need neighbouring data that a bare window does not have. So each window is read with a **400 bp halo** on each side, smoothed, and then cropped back to 25,000. The halo comes from the stored array, never from padding: padding would reintroduce the very edge artefact the halo exists to remove.

Gaussian smoothing reaches $4\sigma = 120$ bp and the running median reaches 187 bp, both comfortably inside 400. Whittaker–Eilers has no finite reach, so its influence never becomes exactly zero — but the residual it leaves is far below the precision at which the counts are stored, and is therefore invisible in practice.

3.4 Validity

A position is *valid* if it is mappable and no smoothing kernel that produced it touched an unmappable base. In practice the mappability mask is grown by 120 bp before use, matching the Gaussian's reach. Positions within 400 bp of a chromosome end have no real halo and are excluded rather than repaired.

Validity is recorded per position and per window. It is currently used for *reporting*, not in the loss.

3.5 Splitting

Chromosomes are assigned whole to one split, so no window can appear on both sides:

train	chr1–6, chr10–22
val	chr7
test	chr8, chr9

4 The model

A 1D U-Net: compress the signal three times, then expand it back, carrying detail across at each level.

4.1 Shape

stage	**length**	**channels**
input	L	6
encoder 1	$L/2$	64
encoder 2	$L/4$	128
encoder 3	$L/8$	256
bottleneck	$L/8$	256
decoder 3–1	back to L	$256 \rightarrow 128 \rightarrow 64$
output	L	1

Each stage halves the length and doubles the channels, so capacity stays roughly constant while the view widens. Skip connections pass each encoder output to the matching decoder, so fine detail is not lost in the compression. The output layer is a 1×1 convolution with no activation: the target is centred near zero and can be negative.

4.2 Inside a block

Every stage contains a residual block of four convolutions, kernel width 7, with dilations 1, 2, 4 and 8. Dilation lets a block see 6, 12, 24 and 48 bp without extra parameters — local shape, transcription-factor footprint, and sub-nucleosomal scale.

Two smaller choices matter. **Reflect padding** rather than zeros, because a coverage window has no natural boundary and zero padding would look like an abrupt loss of coverage. **GroupNorm** rather than BatchNorm, because the batch is small when each example is 25,000 long.

The purely convolutional receptive field is 2,073 bp — about ten nucleosomes.

4.3 Three variants

variant	**description**	**parameters**
V1	coverage and sequence in one shared first convolution	8,517,953
V2	coverage only; the sequence ablation	8,515,393
V3	separate stream per input, joined before the encoder	8,545,665

V3 is the variant in current use. Coverage passes through its own 1D convolution; sequence passes through a 2D convolution spanning all five bases at once and 11 bp along the genome — one full turn of the DNA helix, the periodicity at which sequence influences nucleosome positioning. Each stream produces 32 channels, and the two are concatenated to 64 before the encoder.

The point of the split is that one shared convolution has to serve a continuous count and a categorical one-hot code at the same time; two streams let each be read in its own terms.

5 Loss and training

The loss is **mean squared error** between the reconstruction and the deep track, and nothing else.

Pearson correlation is computed on validation but carries no weight in the objective. It is reported because MSE and correlation can move apart: a model can lower MSE by shrinking its output toward zero without improving the shape at all. Watching both makes that visible.

optimiser	AdamW, learning rate 10^{-3}, weight decay 10^{-2}
schedule	halve the rate after 7 epochs without improvement, floor 10^{-5}
clipping	gradient norm 1.0
batch	16 windows
epochs	40
seed	42

The gradient norm is logged *before* clipping. If it sits near 1.0 the clip is firing constantly and acting as a hidden reduction in learning rate rather than as a safety net.

The validity mask is available to the loss but switched off by default. Using it is a change to the objective, and mixing that with a change to the data would leave neither attributable.

6 Evaluation

Evaluation runs on the held-out chromosomes, one window at a time.

6.1 The comparison that matters

Every metric is reported twice: once for the reconstruction, and once for the **raw undersampled input** against the same target.

This is the point of the evaluation. A reconstruction correlating at 0.57 with the truth means nothing on its own — what matters is what the input already achieved without any model. The gap between the two is the result.

6.2 Metrics

Pearson correlation measures agreement in shape and ignores amplitude.

Mean squared error measures agreement including amplitude.

Peak distance asks the biological question directly: are the nucleosomes in the right places? Peaks are called with a frozen peak caller, and each peak is matched to the nearest peak in the other track.

That last measurement has a trap. Matching each *true* peak to the nearest predicted one rewards predicting more peaks — a track with a peak every few bases scores perfectly. So the distance is measured in both directions:

forward	one distance per true peak; asks *were the real peaks found?*
reverse	one distance per predicted peak; asks *are the predictions real?*

Peak counts for all three tracks are reported alongside. An improvement should only be quoted if it holds in both directions.

A Wilcoxon signed-rank test compares the two sets of peak distances window by window. Paired, because each window supplies both arms.

7 Output

Training writes learning curves, the three best checkpoints by validation loss, and the exact configuration that produced them. Each checkpoint carries its own hyperparameters, so it can be reloaded without external notes.

Evaluation writes:

`test_summary.txt`	the headline numbers
`test_signal_metrics.csv`	per window: MSE, both correlations, valid fraction
`test_peak_distances.csv`	per window: peak counts and both distances
`predictions.tsv.gz`	the reconstructed track, as bedGraph
`ground_truth.tsv.gz`	the deep track, same format

The per-window files are what allow the result to be examined rather than merely quoted — for example, split by mappability.

8 Current results

Held-out chromosomes 8 and 9, sample BH01 (healthy), 5× input:

	reconstruction	**raw input**
Pearson r	0.5729	0.4234
peak distance, forward	22 bp	34 bp
peak distance, reverse	23 bp	31 bp
peaks called (median)	126	115

The true track calls 122 peaks. Windows scored: 9,723. The reconstruction beats its input in 98.8% of windows, and wins in *both* peak directions, so the improvement is not an artefact of calling more peaks.

Performance depends on how mappable the region is:

unmappable fraction	**windows**	**recon r**	**input r**	**gain**
<1%	3,110	0.590	0.410	+0.179
1–10%	4,997	0.584	0.419	+0.166
10–50%	1,375	0.563	0.442	+0.126
>50%	241	0.262	0.651	−0.358

The last row is worth reading carefully. Where more than half the window is unmappable, the *input* appears to correlate better than anywhere else. That is not signal: both coverage levels contain the same read-pileup artefacts, so they agree with each other. The model does not reproduce those artefacts and scores poorly there — which is evidence that what it learned is nucleosome structure rather than mapping error.

9 Known limitations

- The validity mask is not used in the loss. It is available and switched off.

- Windows are kept if any position is valid, so a few almost entirely unmappable windows reach the test set.
- An earlier, inherited data format used different window boundaries and a different position convention. Results from the two cannot be compared window by window; they agree at the level of summary metrics.
- Reconstructed amplitude is systematically smaller than the truth. This is the expected behaviour of a squared-error objective under uncertainty, but it means amplitudes are not comparable between samples of different depth.